Percentages:

1. The price of a laptop increases by 20% and later decreases by 15%. What is the overall percentage change?
2. A town's population decreases by 12% in one year and increases by 15% in the next year. Find the net percentage change.
3. A product undergoes successive price changes of +25%, −20%, and +10%. Find the overall percentage change.
4. A person's salary is increased by 25%. By what percentage should it now be reduced to restore the original salary?
5. A student scored 72% and failed by 21 marks. Another student scored 81% and secured 15 marks more than the passing marks. Find the maximum marks.
6. In an examination, 35% of the students failed in Mathematics, 20% failed in English, and 12% failed in both subjects. What percentage of students passed both subjects?
7. A city's population increases by 8% every year. If its current population is 54,000, what will be the population after two years?
8. A shopkeeper allows a 15% discount and still earns a profit of 19%. By what percentage was the article marked above its cost price?
9. In an election, Candidate A secured 54% of the valid votes and defeated Candidate B by 4,800 votes. Find the total number of valid votes.
10. A person's income increases by 25% while expenditure increases by 20%. If his savings increase by 50%, what was the original expenditure as a percentage of his income?
11. Water is added to milk so that milk constitutes 80% of the mixture. What percentage of the resulting mixture is water?
12. In a company, 40% of the employees are women. Among the women, 30% are managers, while among the men, 20% are managers. What percentage of the employees are managers?
13. A factory increases its production by 25%, but later production falls by 16% due to machinery failure. Find the net percentage change in production.
14. A batsman's average increases from 48 to 54 after one innings. By what percentage has his average increased?
15. A trader marks an article 40% above its cost price and allows a discount of 15%. Find his profit percentage.
16. Out of all registered voters, 12% did not vote. Candidate X secured 55% of the votes polled. What percentage of the registered voters voted for Candidate X?
17. After receiving a 20% salary hike, Rohan's salary became ₹54,000. Find his original salary.

18. A family spends 35% of its income on food, 20% on rent, and 15% on education. What percentage of its income is saved?
19. A shop offers two successive discounts of 20% and 10%. Find the equivalent single discount.
20. A car increases its speed by 20%. By what percentage is the travel time reduced?
21. The length of a rectangle increases by 25% while its breadth decreases by 20%. Find the percentage change in its area.
22. Boys constitute 60% of a class. If the number of girls increases by 25% while the number of boys remains unchanged, what percentage of the class will be girls now?
23. The price of sugar increases by 25%. By what percentage should its consumption be reduced so that the total expenditure remains unchanged?
24. A company increases its dividend from 12% to 15%. Find the percentage increase in dividend.
25. A water tank is 80% full. If one-fourth of the water is removed, what percentage of the tank remains filled?
26. In a school, 60% of the students are boys. If 25% of the boys and 20% of the girls participate in sports, what percentage of the students participate in sports?
27. Sales increased by 30% in the first quarter but decreased by 20% in the second quarter. Find the net percentage change in sales.
28. A person's income decreases by 15%. By what percentage should his expenditure decrease to keep his savings unchanged if originally his expenditure was 80% of his income?
29. In a company, 40% of the employees earn more than ₹50,000. Of them, 30% are women. Among the remaining employees, 20% are women. What percentage of the employees are women?
30. A machine's efficiency increases by 10% every year. What is the effective percentage increase in efficiency after two years?
31. The price of an article is increased by **x%** and then decreased by **x%**. The final price becomes 9% less than the original price. Find **x**.
32. In a college, 48% of the students study Mathematics, 56% study Physics, and 22% study neither subject. What percentage of the students study both Mathematics and Physics?
33. A company's profit increases by 25% while its expenditure increases by 20%. If originally the profit was 15% of the expenditure, find the new profit as a percentage of the new expenditure.
34. A trader marks an article 60% above its cost price and offers two successive discounts of 20% and **x%**. If he still earns a profit of 12%, find **x**.
35. The population of a city grows by 20% every year while 5% of the population migrates out annually. Find the effective annual percentage growth.
36. The price of petrol increases by 20%. By what percentage should consumption be reduced so that the total expenditure increases by only 8%?

37. In an election, 20% of the registered voters did not vote. Of the votes cast, 8% were invalid. Candidate A secured 60% of the valid votes. If Candidate A received 55,200 votes, find the total number of registered voters.
38. An examination consists of three sections carrying equal marks. Student A scores 75%, 80%, and 65% in the three sections respectively. Student B scores 70%, 78%, and 74%. By what percentage does Student A's overall score exceed Student B's?
39. A mixture contains milk and water in the ratio 7:3. Twenty percent of the mixture is removed and replaced with pure milk. What percentage of milk is present in the final mixture?
40. A company's workforce consists of 45% engineers, 35% analysts, and the remaining managers. Among the engineers, 60% are men. Among the analysts, 40% are women. Among the managers, 70% are men. What percentage of the total workforce are women?

Profit, Loss and Discount:

1. A trader purchases an article for ₹2,400 and sells it at a profit of 18%. Find the selling price.
2. A shopkeeper sells an article for ₹1,920 after giving a 20% discount on the marked price. If he earns a profit of 20%, find the marked price.
3. An article is sold at a loss of 15%. Had it been sold for ₹510 more, there would have been a profit of 10%. Find the cost price.
4. A trader marks an article 40% above its cost price and offers a discount of 15%. Find the profit percentage.
5. A shopkeeper marks an article 50% above the cost price and offers successive discounts of 20% and 10%. Find the profit percentage.
6. A trader sells two articles at ₹1,500 each. On one he gains 20%, while on the other he loses 20%. Find his overall profit or loss percentage.
7. A merchant earns a profit of 25% by selling an article. If the cost price had been 10% more and the selling price remained the same, what would have been his profit percentage?
8. By selling an article for ₹3,600, a trader gains 20%. Find the cost price.
9. A shopkeeper purchases 80 calculators for ₹48,000. He sells 60 of them at a profit of 25%. At what profit percentage should he sell the remaining calculators to earn an overall profit of 30%?
10. A trader sells an article at a loss of 12%. If the cost price is ₹2,500, find the selling price.
11. A retailer marks an article 60% above the cost price and offers a 25% discount. Find the profit percentage.

12. A trader purchases an article for ₹4,500 and spends ₹500 on transportation and packaging. He sells it for ₹6,000. Find the profit percentage.
13. An article is marked 30% above its cost price. What single discount should be offered so that the trader earns exactly 10% profit?
14. A shopkeeper offers successive discounts of 15% and 20%. Find the equivalent single discount.
15. A customer gets a single discount of 32%. Find two successive discounts that are equivalent to it.
16. A dealer marks an article 80% above the cost price and allows two successive discounts of 20% and 15%. Find his profit percentage.
17. A trader purchases an article for ₹800 and wants to earn a profit of 35% after offering a discount of 10%. Find the marked price.
18. A trader marks an article 25% above cost price but incurs a loss of 10%. Find the discount percentage offered.
19. An article is sold at a profit of 20%. If both the cost price and selling price increase by ₹300, the profit becomes 25%. Find the original cost price.
20. A shopkeeper earns a profit of 12%. If he had purchased the article for 10% less and sold it at the same price, what would have been his profit percentage?
21. A dishonest shopkeeper claims to sell goods at cost price but uses a weight of only 900 g instead of 1 kg. Find his profit percentage.
22. A trader uses a 950 g weight instead of 1 kg and also charges 5% above the actual cost price. Find his overall profit percentage.
23. A shopkeeper marks his goods at cost price but uses a false weight of 800 g instead of 1 kg. Find his profit percentage.
24. A dishonest merchant uses a 960 g weight instead of 1 kg and sells the goods at a profit of 20%. Find his effective profit percentage.
25. A trader uses a false weight of 850 g instead of 1 kg while offering a 10% discount on the marked price. If the marked price is 40% above the cost price, find his effective profit percentage.
26. A dealer buys 100 kg of rice at ₹60 per kg but sells it using a 950 g weight while charging ₹66 per kg. Find his profit percentage.
27. A trader mixes 20 kg of inferior rice costing ₹40 per kg with 30 kg of superior rice costing ₹60 per kg. If he sells the mixture at ₹65 per kg, find the profit percentage.
28. A merchant buys an article at a discount of 20% on the listed wholesale price and sells it at the listed price. Find his profit percentage.
29. A trader marks an article 50% above cost price. What discount should he offer so that he earns a profit of 20%?
30. A dealer marks an article 75% above cost price and allows successive discounts of 10% and 20%. Find his profit percentage.

31. A trader purchases two articles for a total of ₹8,000. He sells one at a profit of 20% and the other at a loss of 10%. If the overall profit is 8%, find the cost price of each article.
32. A shopkeeper sells an article for ₹2,340 after allowing a discount of 10%. If his profit is 30%, find the cost price.
33. An article is sold for ₹1,840 at a profit of 15%. Had it been sold for ₹160 less, what would have been the profit or loss percentage?
34. A trader buys an article for ₹1,500 and marks it 60% above the cost price. What discount should he offer to earn exactly 28% profit?
35. A wholesaler sells goods to a retailer at a profit of 20%. The retailer marks them 25% above his purchase price and offers a discount of 10%. Find the retailer's profit percentage and the overall profit percentage with respect to the wholesaler's cost price.
36. A trader purchases an article for ₹4,000. He sells it at a profit of 25%. The customer immediately resells it at a profit of 20%. By what percentage is the final selling price more than the original cost price?
37. A shopkeeper marks an article 100% above its cost price and offers three successive discounts of 10%, 10%, and 20%. Find the profit percentage.
38. A dealer uses a false weight of 900 g instead of 1 kg and also mixes 20% sand with sugar. If he sells the mixture at the cost price of pure sugar, find his effective profit percentage.
39. A trader buys 200 notebooks at ₹48 each. During transportation, 10 notebooks are damaged and cannot be sold. At what price per notebook should he sell the remaining notebooks to earn an overall profit of 25%?
40. A shopkeeper offers a festival discount of 15% followed by an additional cashback equivalent to 5% of the discounted price. If the marked price is 50% above the cost price, find the effective profit percentage earned by the shopkeeper.

Averages:

1. The average age of 12 students in a class is 18 years. If the teacher's age is included, the average becomes 21 years. Find the teacher's age.
2. The average of 25 numbers is 48. If one number is excluded, the average becomes 47. Find the excluded number.
3. The average marks of 40 students in a class is 62. If the marks of one student were mistakenly entered as 54 instead of 84, find the correct average.
4. The average salary of 18 employees is ₹42,000. If two new employees with salaries of ₹36,000 and ₹48,000 join the company, find the new average salary.
5. The average of five consecutive even numbers is 36. Find the largest number.
6. The average of seven consecutive odd numbers is 51. Find the smallest number.

7. The average of the first 20 natural numbers is equal to the average of which two middle numbers?
8. The average weight of 15 boys is 52 kg. If one boy weighing 60 kg leaves and another weighing 68 kg joins, find the new average weight.
9. The average score of a batsman after 20 innings is 48 runs. In the 21st innings, he scores 72 runs. Find his new average.
10. The average marks of boys in a class is 72 and that of girls is 68. If there are 18 boys and 12 girls, find the average marks of the entire class.
11. The average age of a family of 6 members is 28 years. If the youngest member is 8 years old, find the average age of the remaining members.
12. The average of 15 numbers is 84. If each number is increased by 6, what will be the new average?
13. The average temperature for Monday to Friday was 30°C. If the temperatures on Saturday and Sunday were 32°C and 34°C respectively, find the average temperature for the entire week.
14. The average of three numbers is 45. If two of them are 36 and 54, find the third number.
15. The average of 40 observations is 72. One observation was wrongly recorded as 96 instead of 69. Find the correct average.
16. The average monthly expenditure of a family for the first 8 months is ₹28,000 and for the last 4 months is ₹35,000. Find the average monthly expenditure for the entire year.
17. The average of 12 numbers is 36. If one number is replaced by another number 24 greater than it, what will be the new average?
18. The average age of 20 students is 16 years. If the teacher's age is included, the average increases by 1 year. Find the teacher's age.
19. The average of 9 numbers is 64. If the average of the first 5 numbers is 60, find the average of the remaining numbers.
20. The average runs scored by a cricketer in 15 innings is 42. If his average after the next 5 innings becomes 48, find the total runs scored in the last 5 innings.
21. The average of 30 students is 58. If the average of boys is 60 and the average of girls is 55, and there are 18 boys, find the number of girls.
22. The average of four consecutive integers is 37.5. Find the integers.
23. The average age of A and B is 24 years. The average age of B and C is 28 years, and the average age of A and C is 26 years. Find the age of B.
24. The average of 16 numbers is 45. If four more numbers having an average of 60 are included, find the new average.
25. The average marks of 50 students is 64. If the highest and lowest marks are removed, the average of the remaining students becomes 65. If the lowest mark is 22, find the highest mark.

26. The average of 11 numbers is 55. If the first five numbers have an average of 48 and the last five numbers have an average of 60, find the middle number.
27. The average speed of a car while traveling from City A to City B is 60 km/h and while returning is 40 km/h. Find the average speed for the entire journey.
28. The average age of 10 employees is 32 years. Five employees with an average age of 28 years leave the company and five new employees with an average age of 36 years join. Find the new average age.
29. The average of 18 numbers is 72. If two numbers, 84 and 96, are removed, find the average of the remaining numbers.
30. A student scores an average of 68 marks in six subjects. How many marks must he score in the seventh subject so that his overall average becomes 72?
31. The average age of 12 members of a committee is 38 years. When the chairman is included, the average age increases by 2 years. Find the chairman's age.
32. The average age of a committee of 20 members is 42 years. If two members aged 36 years and 48 years resign and two new members aged 45 years and 51 years join, find the new average age of the committee.
33. The average age of a committee is 35 years. When a new member joins, the average age becomes 34.5 years. If the committee originally had 18 members, find the age of the new member.
34. The average age of 16 committee members is 40 years. One member aged 52 years retires and is replaced by another member. As a result, the average age of the committee decreases by 1 year. Find the age of the new member.
35. The average age of the members of a committee is 44 years. If the average age of 8 male members is 46 years and the average age of 12 female members is 42 years, find the average age of the entire committee.

Ratios and Proportions:

1. The ratio of the ages of A and B is 5:7. Five years later, the ratio becomes 3:4. Find their present ages.
2. The ratio of the incomes of Ravi and Mohan is 7:9, while the ratio of their expenditures is 5:7. If each saves ₹12,000 per month, find Ravi's monthly income.
3. Two numbers are in the ratio 4:9. If 15 is added to each number, the ratio becomes 7:12. Find the numbers.
4. The ratio of the present ages of a father and his son is 9:2. Ten years ago, the ratio was 7:1. Find their present ages.
5. The salaries of A, B, and C are in the ratio 3:5:7. If C earns ₹24,000 more than A, find B's salary.
6. Three numbers are in the ratio 2:3:5. If their sum is 360, find the largest number.

7. A sum of ₹84,000 is divided among A, B, and C in the ratio 5:7:9. Find the share received by B.
8. The ratio of boys to girls in a class is 7:5. If there are 48 students in the class, how many girls are there?
9. A mixture contains milk and water in the ratio 5:2. If 9 litres of water are added, the ratio becomes 5:5. Find the initial quantity of the mixture.
10. The ratio of two numbers is 11:13. If their HCF is 17, find the larger number.
11. The ratio of two numbers is 8:11. If their LCM is 1056, find the two numbers.
12. The ratio of the speeds of two cars is 4:5. If the first car takes 25 minutes more than the second car to cover the same distance, and the second car takes 100 minutes, find the time taken by the first car.
13. A bag contains red, blue, and green balls in the ratio 3:5:7. If 12 red balls are added, the ratio becomes 5:5:7. Find the original number of balls.
14. The ratio of the monthly incomes of A and B is 8:11. Their expenditures are in the ratio 5:7. If each saves ₹18,000 per month, find the income of B.
15. A sum is divided between A and B in the ratio 4:7. If A receives ₹9,000 less than B, find the total sum.
16. The ratio of the lengths of two trains is 5:7 and the ratio of their speeds is 6:5. If the first train crosses a pole in 30 seconds, how long will the second train take to cross the same pole?
17. The ratio of the capacities of two tanks is 7:9. If Tank A contains 140 litres of water, how much water must be added to Tank A so that the quantity of water in the two tanks is in the ratio 5:6, assuming Tank B is full?
18. Two vessels contain milk and water in the ratios 4:1 and 7:3 respectively. Equal quantities are taken from each vessel and mixed together. Find the ratio of milk to water in the final mixture.
19. The ratio of the present ages of A and B is 4:5. Eight years from now, the ratio of their ages will be 6:7. Find their present ages.
20. Three partners invest in a business in the ratio 4:5:6. At the end of the year, they earn a profit of ₹3,60,000. If the profit is shared in the ratio of their investments, find the share of each partner.

Simple and Compound Interest:

1. Find the simple interest on ₹25,000 at 8% per annum for 4 years.
2. A sum amounts to ₹36,000 in 5 years at 10% simple interest. Find the principal.
3. At what rate of simple interest will ₹18,000 amount to ₹23,400 in 6 years?
4. In how many years will a sum double itself at 12.5% per annum simple interest?
5. A sum earns ₹9,600 as simple interest in 8 years at 10% per annum. Find the principal.
6. The simple interest on a certain sum for 7 years is equal to 70% of the principal. Find the annual rate of interest.
7. A sum of money becomes ₹43,200 in 3 years at simple interest. If the rate is 12% per annum, find the principal.
8. The difference between the amount and the principal at simple interest is ₹5,400. If the rate is 9% per annum for 5 years, find the principal.
9. A person borrows ₹75,000 at simple interest. If he pays ₹96,000 after 4 years, find the annual rate of interest.
10. A principal amounts to ₹52,500 in 5 years and ₹57,000 in 8 years at simple interest. Find the principal and the rate of interest.
11. Find the compound amount on ₹40,000 at 10% per annum for 2 years, compounded annually.
12. Find the compound interest on ₹32,000 at 8% per annum for 3 years, compounded annually.
13. A sum becomes ₹66,550 in 2 years at 15% compound interest. Find the principal.
14. A sum doubles itself in 6 years under compound interest. Find the annual rate of interest (compounded annually).
15. Find the compound interest on ₹48,000 at 12% per annum for 2 years when interest is compounded annually.
16. A sum amounts to ₹59,290 in 3 years at 10% compound interest. Find the principal.
17. At what annual rate of compound interest will ₹25,000 amount to ₹36,300 in 4 years?
18. Find the amount obtained on ₹80,000 after 3 years at 5% compound interest, compounded annually.

19. A sum invested at compound interest grows by ₹9,240 in the second year. If the rate of interest is 10% per annum, find the principal.
20. The compound interest earned in 2 years is ₹12,300 and the principal is ₹50,000. Find the annual rate of interest.
21. The difference between the compound interest and simple interest on a sum for 2 years at 10% per annum is ₹250. Find the principal.
22. The difference between the compound interest and simple interest for 3 years at 20% per annum is ₹1,760. Find the principal.
23. Find the difference between the simple interest and compound interest on ₹60,000 at 12% per annum for 2 years.
24. A sum earns ₹8,000 as simple interest and ₹8,320 as compound interest in 2 years at the same annual rate. Find the principal.
25. A bank offers 10% compound interest compounded annually, while another bank offers 10% simple interest. If ₹1,00,000 is invested for 3 years, find the difference in the maturity amounts.
26. A sum is invested for 2 years at 12% per annum. Calculate the difference in the maturity amounts under simple interest and compound interest.
27. A principal invested at compound interest amounts to ₹79,860 in 2 years and ₹87,846 in 3 years. Find the rate of interest and the principal.
28. A person invests equal amounts in two schemes. One offers 12% simple interest and the other offers 12% compound interest, both for 2 years. If the difference in the maturity amounts is ₹432, find the amount invested in each scheme.
29. A loan of ₹2,00,000 is taken at 9% compound interest compounded annually. Find the amount payable after 3 years if no repayment is made during the period.
30. A finance company offers compound interest at 8% per annum for the first two years and 10% per annum for the third year. If ₹50,000 is invested, find the maturity amount at the end of 3 years.

Time and Work

1. A can complete a piece of work in 12 days, while B can complete it in 18 days. In how many days can they complete the work together?
2. A can finish a job in 15 days and B in 20 days. They work together for 5 days, after which A leaves. In how many more days will B finish the remaining work?

3. A, B, and C can complete a work individually in 10, 15, and 30 days respectively. They start working together, but C leaves after 3 days. How many more days will A and B take to finish the remaining work?
4. A is twice as efficient as B. If B alone can complete a work in 24 days, in how many days can they complete it together?
5. A can complete a work in 24 days, B in 16 days, and C in 12 days. They work on alternate days starting with A. In how many days will the work be completed?
6. A can do a work in 18 days and B can do the same work in 24 days. They work together for 6 days, after which B leaves. How many more days will A take to finish the remaining work?
7. Twelve workers can complete a project in 15 days. After working for 5 days, 3 workers leave. How many additional days will the remaining workers take to complete the project?
8. A contractor hires 24 workers who can complete a work in 20 days. After 8 days, 6 more workers join. In how many total days will the work be completed?
9. A can complete a work in 30 days and B in 45 days. They work together for 10 days, after which B leaves. What fraction of the work is still left?
10. A completes 40% of a work in 8 days. At the same rate, how many more days will he take to finish the remaining work?
11. A can complete a work in 20 days. After working for 8 days, he is assisted by B, and together they finish the work in 6 more days. In how many days can B alone complete the work?
12. A and B together complete a work in 9 days. B alone can complete it in 15 days. In how many days can A alone complete the work?
13. Three workers A, B, and C can complete a work in 12, 18, and 36 days respectively. If they work together, what percentage of the work remains after 4 days?
14. If 15 men can complete a work in 24 days, how many men are required to complete the same work in 18 days?
15. A is 50% more efficient than B. If both work together and complete a job in 8 days, how many days will A alone take to complete the work?
16. Pipe A can fill a tank in 12 hours and Pipe B can fill it in 18 hours. How long will they take together to fill the tank?
17. Pipe A fills a tank in 15 hours, while Pipe B empties it in 20 hours. If both pipes are opened together, how long will it take to fill the tank?

18. Two inlet pipes can fill a tank in 10 hours and 15 hours respectively, while an outlet pipe can empty it in 30 hours. If all three pipes are opened together, how long will it take to fill the tank?

19. Pipe A fills a tank in 8 hours. After 3 hours, Pipe B is opened, and together they fill the tank in another 2 hours. In how many hours can Pipe B alone fill the tank?

20. Pipe A can fill a tank in 24 hours and Pipe B can empty it in 36 hours. If both are opened together for 12 hours, what fraction of the tank will be filled?

21. Three pipes A, B, and C can fill a tank in 12, 16, and 24 hours respectively. If all are opened together, how long will they take to fill the tank?

22. Pipe A fills a tank in 20 hours. Pipe B empties it in 30 hours. If Pipe A is opened for 8 hours before Pipe B is opened, how much additional time will be required to fill the tank?

23. Two pipes fill a tank in 18 hours and 24 hours respectively. If they are opened alternately for one hour each, starting with the faster pipe, how long will it take to fill the tank?

24. A leak can empty a full tank in 40 hours. An inlet pipe can fill it in 10 hours. If the leak is present throughout, how long will it take to fill the empty tank?

25. Pipe A fills a tank in 14 hours and Pipe B fills it in 21 hours. After both pipes have been opened for 4 hours, Pipe B is closed. How much more time will Pipe A require to fill the remaining tank?

26. If 18 workers working 8 hours a day can complete a project in 15 days, how many workers are required to complete the same project in 12 days by working 10 hours a day?

27. Twenty-four workers working 6 hours a day can manufacture 3,600 units in 15 days. How many workers working 8 hours a day are required to manufacture 6,400 units in 20 days?

28. Eight machines can produce 2,400 bolts in 10 hours. How many machines are required to produce 5,400 bolts in 15 hours, assuming all machines work at the same rate?

29. Fifteen men can complete a work in 18 days by working 9 hours a day. How many men are required to complete twice the work in 15 days by working 12 hours a day?

30. Twenty workers working 7 hours a day complete 70% of a project in 12 days. How many additional workers are required to finish the remaining work in the next 4 days if they continue working 7 hours a day?

31. Sixteen workers working 8 hours a day can construct 240 metres of road in 15 days. How many workers working 10 hours a day are needed to construct 500 metres of the same road in 20 days?
32. Twelve machines working 9 hours a day manufacture 5,400 components in 18 days. How many machines working 6 hours a day are needed to manufacture 7,200 components in 24 days?
33. Thirty workers working 8 hours a day can complete a project in 24 days. After 8 days, 6 workers leave the project. By how many days will the completion of the project be delayed?
34. Forty workers working 6 hours a day can complete a contract in 18 days. If they work only 5 hours a day, how many additional workers are required to complete the contract in the same time?
35. Twenty men or thirty women can complete a work in 24 days. In how many days can 10 men and 15 women complete the same work together?

Ages:

1. The present ages of A and B are in the ratio 5:7. Five years from now, their ages will be in the ratio 3:4. Find their present ages.
2. The present age of a father is three times the age of his son. After 12 years, the father will be twice as old as the son. Find their present ages.
3. The average age of a family of 6 members is 32 years. A child is born, and after 4 years, the average age of the family becomes 31 years. Find the present age of the newborn child.
4. Ten years ago, the ratio of the ages of A and B was 3:5. Ten years from now, the ratio will be 5:7. Find their present ages.
5. The sum of the present ages of a father and his daughter is 60 years. Five years ago, the father's age was five times the daughter's age. Find their present ages.
6. The ratio of the present ages of A, B, and C is 4:5:6. Eight years from now, the ratio of their ages will be 6:7:8. Find their present ages.
7. The present age of a mother is four times the age of her son. Eight years ago, the mother was six times as old as her son. Find their present ages.
8. The average age of A, B, and C is 28 years. If the average age of A and B is 26 years and the average age of B and C is 30 years, find the age of B.

9. The present age of a father is 24 years more than that of his son. Six years from now, the father's age will be twice the son's age. Find their present ages.
10. A is 6 years older than B, and B is 4 years older than C. If the sum of their present ages is 54 years, find the present ages of A, B, and C.

Partnership:

1. A and B invest ₹80,000 and ₹1,20,000 respectively in a business. At the end of the year, they earn a profit of ₹50,000. Find the share of each partner.
2. A, B, and C invest in the ratio 3:5:7. If the total profit at the end of the year is ₹2,40,000, find the share of each partner.
3. A invests ₹60,000 for 12 months, while B invests ₹90,000 for 8 months. Find the ratio in which the profit should be divided.
4. A starts a business with ₹1,50,000. After 4 months, B joins with ₹2,00,000. At the end of one year, the total profit is ₹1,92,000. Find the share of each partner.
5. A and B invest ₹80,000 and ₹1,20,000 respectively. After 6 months, A withdraws ₹20,000 while B adds ₹40,000. If the annual profit is ₹78,000, find each partner's share.
6. A, B, and C invest ₹50,000, ₹75,000, and ₹1,00,000 respectively. After 4 months, C withdraws half of his investment. If the profit at the end of the year is ₹1,26,000, find the share of each partner.
7. A and B invest in the ratio 5:7. After 8 months, A doubles his investment. If the total annual profit is ₹72,000, find the share of A.
8. A and B start a business with equal investments. After 5 months, A withdraws 20% of his capital while B adds 40% to his capital. If the total profit at the end of the year is ₹84,000, find the profit share of B.
9. A, B, and C invest in the ratio 4:6:5. B joins the business after 3 months, while C joins after 6 months. Find the ratio in which the annual profit should be divided.
10. A invests ₹2,40,000 and B invests ₹1,80,000. If A is also paid a salary of ₹6,000 per month for managing the business, and the remaining annual profit is ₹2,16,000, find A's total earnings.
11. A and B invest in the ratio 2:3. If A receives ₹24,000 less profit than B, find the total profit earned by the business.

12. Three partners invest in the ratio 5:4:3. If the difference between the highest and lowest profit shares is ₹60,000, find the total profit.
13. A invests ₹1,20,000 for the entire year. B invests ₹90,000 but withdraws his investment after 8 months. If the total profit is ₹1,10,000, find B's share.
14. A and B started a business by investing ₹2,00,000 and ₹3,00,000 respectively. After 4 months, C joins with ₹4,00,000. At the end of the year, the business earns a profit of ₹3,60,000. Find the share of C.
15. A, B, and C invest ₹90,000, ₹60,000, and ₹1,20,000 respectively. After 6 months, A withdraws one-third of his capital, while B doubles his investment. Find the ratio in which the profit should be distributed.
16. Two partners invest in the ratio 7:5. After 3 months, the first partner withdraws one-fourth of his investment, while the second partner increases his investment by 20%. Find the ratio of their profit shares at the end of the year.
17. A and B invest ₹1,00,000 and ₹1,50,000 respectively. After 4 months, A adds ₹50,000 to his investment, while B withdraws ₹30,000. If the total annual profit is ₹1,80,000, find the profit share of A.
18. Four partners A, B, C, and D invest in the ratio 2:3:5:6. If the business earns a profit of ₹4,80,000, find the share of partner C.
19. A invests twice as much as B, while B invests three times as much as C. If the total annual profit is ₹1,80,000, find the share of each partner.
20. A, B, and C invest ₹1,00,000, ₹1,50,000, and ₹2,00,000 respectively. After 3 months, A withdraws 25% of his investment, B adds ₹50,000, and C withdraws ₹1,00,000. If the total profit at the end of the year is ₹3,60,000, find the profit share of each partner.

Time, Speed and Distance:

1. A car travels 360 km at a speed of 60 km/h and returns at a speed of 90 km/h. Find the average speed for the entire journey.
2. A train travels at 72 km/h. How much time will it take to cover a distance of 540 metres?
3. Two cities are 480 km apart. A car travels from City A to City B at 60 km/h and returns at 80 km/h. Find the total time taken for the round trip.
4. A person covers one-third of a journey at 40 km/h, one-third at 60 km/h, and the remaining one-third at 80 km/h. Find his average speed for the entire journey.

5. A cyclist increases his speed by 25% and reaches his destination 24 minutes earlier than usual. Find the original time taken.
6. A person travels half the distance at 50 km/h and the remaining half at 75 km/h. Find the average speed.
7. A car covers the first 180 km at 45 km/h and the next 240 km at 60 km/h. Find the average speed for the entire journey.
8. Walking at 5 km/h, a person reaches his office 12 minutes late. Walking at 6 km/h, he reaches 8 minutes early. Find the distance between his home and office.
9. A train running at 54 km/h crosses a platform 180 metres long in 30 seconds. Find the length of the train.
10. A person reduces his speed by 20% and consequently takes 30 minutes more to cover a certain distance. Find the original time taken.
11. A train 180 metres long crosses a pole in 12 seconds. Find its speed in km/h.
12. A train 240 metres long crosses a platform 360 metres long in 30 seconds. Find the speed of the train.
13. Two trains of lengths 180 metres and 220 metres are moving in the same direction at 54 km/h and 72 km/h respectively. How long will the faster train take to completely overtake the slower train?
14. Two trains of lengths 240 metres and 160 metres are moving in opposite directions at speeds of 72 km/h and 54 km/h respectively. Find the time taken for them to completely cross each other.
15. A train crosses a man walking at 6 km/h in the same direction in 18 seconds. The length of the train is 180 metres. Find the speed of the train.
16. A train crosses another train of equal length moving in the opposite direction in 10 seconds. Both trains are 150 metres long. If one train is moving at 54 km/h, find the speed of the other train.
17. A train passes a pole in 15 seconds and passes a platform 240 metres long in 27 seconds. Find the length and speed of the train.
18. Two trains moving in opposite directions at 90 km/h and 54 km/h take 12 seconds to cross each other. If one train is 180 metres long, find the length of the other train.
19. A boat travels 48 km downstream in 3 hours and the same distance upstream in 4 hours. Find the speed of the boat in still water and the speed of the stream.
20. A boat's speed in still water is 18 km/h, and the speed of the stream is 3 km/h. Find the time taken to travel 84 km upstream.

21. A boat covers 90 km downstream in 5 hours and returns upstream in 7.5 hours. Find the speed of the stream.
22. The speed of a boat in still water is twice the speed of the stream. If the boat takes 4 hours to travel 48 km downstream, find the speed of the stream.
23. A boat travels 72 km downstream in 4 hours and 60 km upstream in 5 hours. Find the speed of the boat in still water.
24. A man can row a boat at 12 km/h in still water. If the stream flows at 4 km/h, find the time taken to row 64 km downstream and return.
25. A boat takes 6 hours to travel downstream from A to B and 9 hours to return upstream. If the distance between A and B is 72 km, find the speed of the boat in still water.
26. In a 400-metre race, A beats B by 40 metres. By how many metres will A beat B in a 600-metre race?
27. In a race of 500 metres, A beats B by 5 seconds. If A's speed is 10 m/s, find B's speed.
28. In a 300-metre race, A beats B by 30 metres and B beats C by 20 metres. By how many metres will A beat C in a 300-metre race?
29. In a race, A gives B a start of 40 metres in a 200-metre race and still wins by 20 metres. Find the ratio of their speeds.
30. A and B run a race of 600 metres. A can give B a start of 75 metres. By what distance will A beat B in a race of 800 metres?
31. Two cars start simultaneously from two cities 540 km apart and travel towards each other at speeds of 60 km/h and 75 km/h. After how many hours will they meet?
32. A man walks to his office at 4 km/h and returns by the same route at 6 km/h. Find his average speed for the round trip.
33. A train running at 90 km/h crosses a bridge 360 metres long in 36 seconds. Find the length of the train.
34. Two cyclists start from the same point in opposite directions at speeds of 18 km/h and 24 km/h. How far apart will they be after 2 hours 30 minutes?
35. A person covers a journey by car at 60 km/h and returns by bus at 40 km/h. If the total journey takes 10 hours, find the one-way distance.
36. A train leaves Station A at 8:00 AM at a speed of 60 km/h. Another train leaves the same station at 10:00 AM at a speed of 90 km/h. At what time will the second train catch up with the first?

37. A train 210 metres long crosses a platform in 24 seconds. It crosses a pole in 14 seconds. Find the length of the platform.
38. A boat travels 54 km downstream in 3 hours and 36 km upstream in 3 hours. Find the speed of the stream.
39. Two trains 180 metres and 240 metres long are moving in the same direction at 72 km/h and 54 km/h respectively. How long will it take for the faster train to completely overtake the slower train?
40. A person walks at 5 km/h and reaches a railway station 10 minutes late. If he walks at 6 km/h, he reaches 5 minutes early. Find the distance between his home and the station.

Probability

1. A bag contains 8 red, 6 blue, and 4 green balls. One ball is drawn at random. Find the probability that it is either red or green.
2. Two fair dice are rolled simultaneously. Find the probability that the sum of the numbers obtained is 9.
3. A card is drawn from a standard deck of 52 cards. Find the probability that the card is a face card or a heart.
4. Three coins are tossed simultaneously. Find the probability of getting exactly two heads.
5. A bag contains 5 white, 7 black, and 8 yellow balls. Two balls are drawn one after another without replacement. Find the probability that both are black.
6. A number is selected at random from 1 to 100. Find the probability that it is divisible by both 3 and 5.
7. Two cards are drawn from a deck of 52 cards without replacement. Find the probability that both are kings.
8. A letter is chosen at random from the word **PROBABILITY**. Find the probability that the selected letter is a vowel.
9. Three fair dice are rolled together. Find the probability that all three show the same number.
10. A bag contains 10 balls numbered 1 to 10. One ball is drawn at random. Find the probability that the number is prime.

11. A box contains 6 defective and 14 non-defective bulbs. Two bulbs are selected randomly without replacement. Find the probability that exactly one is defective.
12. A card is drawn from a well-shuffled deck. Find the probability that it is neither a king nor a queen.
13. A fair die is rolled twice. Find the probability that the product of the two numbers obtained is even.
14. Four coins are tossed simultaneously. Find the probability of getting at least three heads.
15. A committee of 3 members is to be formed from 5 men and 4 women. What is the probability that the committee contains exactly two women?
16. A student answers a multiple-choice question with four options by guessing. What is the probability that he answers correctly?
17. Two fair dice are rolled. Find the probability that at least one die shows a 6.
18. A bag contains 5 red, 4 blue, and 3 green balls. Three balls are drawn simultaneously. Find the probability that all three are of different colors.
19. A card is drawn from a deck of 52 cards. Find the probability that it is an ace or a spade.
20. Two numbers are chosen randomly from the integers 1 to 20 without replacement. Find the probability that both are even.
21. A box contains 8 good and 2 defective mobile chargers. Three chargers are selected at random without replacement. Find the probability that all three are good.
22. A family has two children. Assuming each child is equally likely to be a boy or a girl, what is the probability that both are girls, given that at least one child is a girl?
23. A fair die is rolled three times. Find the probability of getting exactly one six.
24. A card is drawn from a deck of 52 cards and replaced. Another card is drawn. Find the probability that both cards are queens.
25. A jar contains 12 chocolates, of which 5 are dark chocolates. Three chocolates are selected without replacement. Find the probability that none of them is a dark chocolate.
26. A password consists of a single digit chosen randomly from 0 to 9. What is the probability that the digit is greater than 5?
27. From a group of 8 boys and 6 girls, two students are selected at random. Find the probability that both selected students are girls.

28. A fair coin is tossed five times. Find the probability of getting exactly three heads.
29. A bag contains 4 red, 5 blue, and 6 green marbles. Two marbles are drawn one after another without replacement. Find the probability that the second marble is green.
30. A class has 18 boys and 12 girls. Two students are selected at random. Find the probability that one is a boy and the other is a girl.

Permutation and Combination

1. In how many different ways can the letters of the word **COMPUTER** be arranged?
2. In how many ways can 5 students be selected from a class of 12 students?
3. In how many different ways can 4 books be arranged on a shelf from a collection of 10 different books?
4. A committee of 4 members is to be formed from 7 men and 5 women. In how many ways can this be done?
5. In how many ways can the letters of the word **SUCCESS** be arranged?
6. How many 4-digit numbers can be formed using the digits 1, 2, 3, 4, 5, and 6 without repetition?
7. In how many ways can the letters of the word **ENGINEER** be arranged?
8. A cricket team of 11 players is to be selected from 15 players. In how many ways can the team be selected?
9. In how many ways can a President, Vice-President, and Secretary be chosen from a group of 12 people?
10. How many different 5-digit even numbers can be formed using the digits 1, 2, 3, 4, 5, and 6 without repetition?
11. In how many ways can 6 people sit around a circular table?
12. In how many ways can 8 people sit around a circular table if two particular people must always sit together?
13. In how many ways can the letters of the word **BALLOON** be arranged?
14. From a group of 8 boys and 6 girls, in how many ways can a committee of 5 members be formed consisting of exactly 3 boys and 2 girls?
15. In how many ways can 3 Mathematics books and 4 Physics books be arranged on a shelf if books of the same subject must remain together?

16. How many 6-digit numbers greater than 500000 can be formed using the digits 1, 2, 3, 4, 5, and 6 without repetition?
17. In how many ways can the letters of the word **MISSISSIPPI** be arranged?
18. A team of 6 members is to be selected from 10 men and 8 women such that at least 2 women are included. In how many ways can the team be formed?
19. In how many ways can 5 different prizes be distributed among 8 students if each student can receive at most one prize?
20. In how many ways can 7 people stand in a line if two particular persons are never together?
21. In how many ways can the letters of the word **MATHEMATICS** be arranged?
22. A committee of 5 members is to be formed from 9 men and 7 women such that the committee contains at least 3 women. In how many ways can the committee be formed?
23. In how many ways can 4 boys and 4 girls be seated in a row such that boys and girls sit alternately?
24. How many 5-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6 without repetition?
25. In how many ways can 10 people be seated in a row if three particular persons must always be together?
26. From a group of 12 players, in how many ways can a captain and a vice-captain be selected?
27. In how many ways can the letters of the word **ARRANGEMENT** be arranged?
28. A committee of 6 members is to be formed from 8 men and 7 women such that there are more women than men. In how many ways can the committee be formed?
29. In how many ways can 9 people be seated around a circular table if two particular people must not sit together?
30. A password consists of 4 distinct letters followed by 2 distinct digits. If the letters are chosen from the English alphabet and the digits from 0–9, in how many different passwords can this be formed?

Set Theory and Venn Diagram

1. In a class of 100 students, 60 like Mathematics, 45 like Science, and 25 like both. How many students like neither subject?

2. In a survey of 250 people, 140 like Tea, 120 like Coffee, and 70 like both. How many people like only Tea?
3. In a college, 180 students enrolled in Python, 150 enrolled in Java, and 90 enrolled in both. If there are 300 students in total, how many enrolled in neither course?
4. In a company, 65 employees know Excel, 50 know SQL, and 30 know both. If there are 100 employees, how many know exactly one of the two skills?
5. Among 150 students, 80 play Cricket, 65 play Football, and 35 play both. How many students play only Football?
6. In a survey of 500 households, 320 own a television, 280 own a refrigerator, and 180 own both. How many households own neither appliance?
7. In a coaching institute, 120 students study Quantitative Aptitude, 90 study Logical Reasoning, and 50 study both. If there are 180 students, how many study only Quantitative Aptitude?
8. In a class of 90 students, 55 passed Mathematics, 48 passed Science, and 36 passed both. How many students failed in both subjects?
9. Out of 300 employees, 170 use Windows, 140 use Linux, and 90 use both operating systems. How many employees use only Linux?
10. In a survey, 52% of people like Pizza, 38% like Burgers, and 18% like both. What percentage of people like neither Pizza nor Burgers?
11. In a school, 240 students participate in Music, 180 participate in Dance, and 120 participate in both. If every student participates in at least one activity, find the total number of students.
12. In a group of 200 people, 120 speak English, 90 speak Hindi, and 50 speak both languages. How many speak exactly one language?
13. Out of 150 students, 75 study C++, 60 study Python, and 25 study both. How many students study neither C++ nor Python?
14. In a survey of 400 people, 220 watch Cricket, 180 watch Football, and 110 watch both. How many watch only Cricket?
15. In a batch of 180 students, 100 cleared Aptitude, 90 cleared Coding, and 60 cleared both rounds. How many students cleared exactly one of the two rounds?
16. In a survey of 300 people, 180 use WhatsApp, 150 use Telegram, and 90 use both. If 20 people use neither app, how many use only WhatsApp?

17. Among 250 students, 140 know Python, 110 know Java, and 60 know both. If every student knows at least one programming language, how many know exactly one language?
18. In a college festival, 210 students participated in Singing, 170 in Dancing, and 90 in both events. If 60 students participated in neither event, find the total number of students.
19. In a survey of 500 people, 280 like Online Shopping, 240 like Offline Shopping, and 150 like both. Find the number of people who like exactly one mode of shopping.
20. In a placement drive, 220 students cleared the Aptitude round, 180 cleared the Technical round, and 140 cleared at least one of the two rounds. Determine whether the given data is possible. If not, identify which value(s) must be incorrect and explain why.

Number System:

1). Find the greatest 3-digit number, which is a multiple of 21.

2). Find the smallest 4-digit number, which is a multiple of 12.

3). Which letter should replace the x in the number 6377x99, so that it becomes a multiple of 9.

4). What will be the number of zeroes at the end of the product of the first 100 natural numbers?

5). Find the prime factors of 1200.

6). Find the unit digit of 288^6693.

7). Find the unit digit of 555^555555555.

8). Find the unit digit of 139^489

9). Find the remainder of 19^2200002/23.

10). Find the remainder of 13^18/19.

11). Find the remainder when 11^74 is divided by 63.

12). Find the Remainder of (15 * 18 * 13) / 5

13). What is the number of zeros present at the end of 11 x 12 × 13 × 14 × 15 × 16 × 17 x 19?

14). How many zeros will be present at the end of 5 x 15 × 25 × 35 × 75 * 125?

15). What will be the number of zero at the end of 150!

16). What are the number of factors of 360.

17). Find the sum of factors of 720. In how many ways can 720 be written as product of two factors.

18). Find the square root of 3969.

19). Find the cube root of 9261. Using factorisation.

20). What is the LCM and HCF of 1/2 and 2/3.

21). Find the greatest number which when divides 139 and 167 leaves remainders 4 and 5 respectively.

22). Find the least number which when divided by 8, 16, 20 and 24 leaves remainder 5 in each case.

23). Find the least number which when divided by 8, 12, 20 leaves remainders 5, 9 and 17 respectively.

24). Find the greatest number, which when 31, 51 and 91 leave the same remainder in each case.

Blood Relations:

1. Pointing to a man, Rohan said, "He is the son of my grandfather's only son." How is the man related to Rohan?
2. A woman introduces a man as, "He is the only son of my mother's only daughter." How is the man related to the woman?
3. If A is the brother of B, B is the sister of C, and C is the father of D, how is A related to D?
4. Pointing to a photograph, Neha said, "The woman in the photograph is the daughter of my father's only sister." How is the woman related to Neha?
5. P is the father of Q. R is the wife of P. S is the daughter of R, and T is the brother of S. How is Q related to T?
6. Introducing a woman, Raj said, "She is the daughter of the only son of my grandfather." How is the woman related to Raj?

7. If A is B's mother, C is A's father, D is C's wife, and E is D's son, how is E related to B?
8. Pointing towards a boy, Kiran said, "He is the grandson of the father of my husband." How is the boy related to Kiran?
9. A man said, "My father's brother's only daughter is a doctor." How is the doctor related to the man?
10. If X is the sister of Y, Y is the father of Z, and Z is the brother of P, how is X related to P?
11. A woman says, "The person standing next to me is the son of my husband's father." How is that person related to the woman?
12. Pointing to a girl, Aman said, "She is the daughter of the wife of my father's only son." How is the girl related to Aman?
13. A is the maternal grandfather of B. C is the only daughter of A. D is the husband of C. How is D related to B?
14. If P is the son of Q, Q is the daughter of R, and R is the wife of S, how is S related to P?
15. Looking at a family photograph, Meera said, "The man sitting in the center is the husband of my mother's sister." How is the man related to Meera?
16. Pointing to a man, Ravi said, "His mother is the only daughter of my grandmother." How is the man related to Ravi?
17. A introduces B as the son of the only brother of A's mother. How is B related to A?
18. If A is the father of B, B is the husband of C, C is the mother of D, and E is the brother of B, how is E related to D?
19. Pointing to a woman, Suresh said, "She is the wife of the only son of my father's father." How is the woman related to Suresh?
20. A man said, "I have no siblings. The father of this boy is the son of my father." How is the boy related to the man?

Directions:
means Father of
@ means Mother of
$ means Brother of
% means Sister of
& means Husband of
* means Wife of

1.
If # means Father of, @ means Mother of, $ means Brother of, and % means Sister of, then what is the relation of P to T?
P # Q @ R $ S % T

2.
If # means Father of, @ means Mother of, & means Husband of, and * means Wife of, then how is A related to E?
A & B @ C # D * E

3.
If # means Father of, $ means Brother of, % means Sister of, and @ means Mother of, then how is M related to Q?
M # N $ O @ P % Q

4.
If # means Father of, @ means Mother of, $ means Brother of, % means Sister of, and & means Husband of, then how is X related to V?
X & Y @ Z # W $ V

5.
If # means Father of, @ means Mother of, $ means Brother of, % means Sister of, & means Husband of, and * means Wife of, then how is K related to P?
K # L & M @ N $ O % P

Coding and Decoding:

1. In a certain code language, **COMPUTER** is written as **RFUVQNPD**. How will **KEYBOARD** be written in that code?
2. If **APPLE** is coded as **ELPPA**, how will **ORANGE** be coded using the same pattern?
3. In a certain code, **MANGO** is written as **OCPIQ**. How is **GRAPE** written in that code?
4. If **DELHI** is coded as **73549** and **HIDE** is coded as **4937**, what is the code for **HELD**?
5. In a certain code language, **PEN** is written as **QFO**. How will **BOOK** be written?
6. If **TRAIN** is coded as **YWFNS**, how will **PLANE** be coded using the same pattern?
7. In a certain language, **987** means **"good morning everyone"**, **764** means **"everyone likes coding"**, and **479** means **"coding is good"**. Which digit represents **coding**?
8. In a certain code language, **MOBILE** is written as **NQDKMF**. How will **LAPTOP** be written?
9. If **123** means **"I love coding"**, **345** means **"coding is fun"**, and **356** means **"fun and learning"**, which number represents **fun**?
10. In a certain code language, each letter is replaced by the letter two positions ahead in the English alphabet. How will **PLACEMENT** be coded?
11. If **GARDEN** is coded as **MFXHIU**, what is the logic used in the coding?

12. In a certain code language, every vowel is replaced by the next vowel and every consonant is replaced by the next consonant. How will **PROJECT** be coded?
13. In a certain code, **TABLE** is written as **UCDNG**. How will **CHAIR** be written?
14. If **ROSE** is coded as **18-15-19-5**, how will **LILY** be coded?
15. In a certain code language, the letters of every word are written in reverse order, and then the entire word is shifted one position to the right. How will **COMPUTER** be coded?
16. If **BAT = 45**, **CAT = 46**, and **DOG = 26**, based on the sum of alphabetical positions, what will be the code for **FISH**?
17. In a certain code language, **HAPPY** is written as **IBQQZ**. How will **SMART** be written?
18. In a certain language, **"Data Structures are Important"** is coded as **ka lo mi na**, **"Important for Placements"** is coded as **na pi su**, and **"Data Analytics"** is coded as **ka re**. What is the code for **Important**?
19. If **MACHINE** is coded as **NZDIOJMF**, how will **NETWORK** be coded using the same pattern?
20. In a certain code language, the first letter of a word is replaced by the last letter, the second by the second last, and so on. How will **PROGRAMMING** be coded?

Direction Sense:

1. A person starts from his home and walks 80 m towards the North. He turns left and walks 60 m. He again turns left and walks 80 m. Finally, he turns right and walks 40 m. How far and in which direction is he from his home?
2. Priya walks 120 m towards the East. She turns right and walks 90 m. She then turns left and walks 50 m. Finally, she turns left and walks 90 m. Determine her final position with respect to the starting point.
3. Rahul starts facing North. He turns 135° clockwise, then 90° anticlockwise, then 45° clockwise. Which direction is he facing now?
4. A person walks 25 m towards the South, turns left and walks 40 m, turns left again and walks 25 m, and finally turns right and walks 15 m. Find the shortest distance from the starting point.
5. Kiran starts from Point A and walks 18 km towards the West. He then turns right and walks 12 km. He again turns right and walks 18 km. In which direction is he from Point A?
6. A man starts facing East. He turns 270° clockwise, then 180° anticlockwise, and finally 90° clockwise. Which direction is he facing now?
7. During sunrise, Amit and Rohit are standing on the top of a hill facing each other. Amit's shadow falls exactly towards Rohit's left. In which direction is Rohit facing?

8. One morning after sunrise, Neha is facing a flagpole. The shadow of the flagpole falls exactly to her right. Which direction is Neha facing?
9. During sunset, P and Q are standing face-to-face. If P's shadow falls exactly towards his left, in which direction is Q facing?
10. A person starts walking towards the Sun early in the morning. He walks 100 m, turns 90° to his left and walks 80 m, then turns 180° and walks 30 m. In which direction is he now from the starting point?
11. At sunset, two mountaineers are standing face-to-face on a hill. One of them is facing East. What is the direction faced by the other mountaineer? Also determine the direction of their shadows.
12. A clock is placed on a table such that at 3:00 PM the hour hand points towards the North. In which direction will the minute hand point at 9:15 PM?
13. A clock is placed on a table such that at 12 noon the minute hand points towards the East. In which direction will the hour hand point at 6:00 PM?
14. Ravi starts facing South-East. He turns 45° clockwise, then 90° anticlockwise, and then 180° clockwise. Which direction is he facing now?
15. If North becomes South-East, South becomes North-West, East becomes South-West, and West becomes North-East, then which direction will North-West become?
16. If East is called North, North is called West, West is called South, and South is called East, then in which direction will a person move if he actually walks towards the West?
17. A man walks 14 m North, then 14 m East, then 14 m South, then 14 m West. Where is he with respect to his starting point?
18. A person walks 35 m towards the East, turns left and walks 24 m, turns left again and walks 35 m, then turns right and walks 18 m. Find the shortest distance from his starting point.
19. A boy starts facing West. He turns left, then left again, then right, then right, and finally turns left. Which direction is he facing?
20. Ramesh starts from his office and walks 150 m North. He turns right and walks 80 m. He then turns right and walks 100 m. Finally, he turns left and walks 60 m. Determine his final position relative to the office.
21. During sunrise, A and B are standing face-to-face. A is facing West. Is the situation possible? If yes, determine the direction faced by B. If not, explain why.
22. A person walks 12 km towards the North. He turns right and walks 5 km. He again turns right and walks 8 km. Then he turns left and walks 7 km. How far and in which direction is he from the starting point?
23. One morning after sunrise, a man notices that his shadow is pointing exactly towards the South-West. Is this situation possible? Justify your answer.

24. A person starts facing North. He makes four consecutive turns of 45°, 90°, 135°, and 90° clockwise. Which direction is he facing at the end?
25. A person starts from Point A, walks 40 m North, 30 m East, 40 m South, 20 m West, 10 m South, and finally 10 m East. Find the shortest distance and direction of the final position from Point A.

Clocks:

1. What is the smaller angle between the hour hand and the minute hand at **7:20**?

2. Find the angle between the hour hand and the minute hand at **5:45 PM**.

3. What is the larger angle between the hands of a clock at **8:25**?

4. At what time between **3 PM and 4 PM** will the hands of a clock coincide?

5. At what time between **6 PM and 7 PM** will the hands of a clock be exactly **180°** apart?

6. At what time between **9 PM and 10 PM** will the hands of the clock make an angle of **90°** for the first time?

7. How many times do the hands of a clock coincide between **12 noon and 12 midnight**?

8. How many times do the hands of a clock form a right angle in **24 hours**?

9. Between **2 PM and 5 PM**, how many times do the hands of a clock form an angle of **90°**?

10. Between **8 AM and 11 AM**, how many times do the hands of a clock form a straight angle (180°)?

11. At what time between **1 PM and 2 PM** will the hands of the clock be **60°** apart?

12. At what time between **10 AM and 11 AM** will the hands of the clock be **120°** apart?

13. At what time between **4 PM and 5 PM** will the hands of the clock be **150°** apart?

14. If the actual time is **4:40**, what time will be shown in the **mirror image** of the clock?

15. A mirror shows the time as **8:35**. What is the actual time?

16. If the mirror image of a clock shows **2:15**, find the correct time.

17. A clock gains **4 minutes every hour**. If it is set correctly at **6:00 AM**, what time will it show when the correct time is **6:00 PM**?

18. A clock loses **3 minutes every hour**. If it is set correctly at **8:00 AM**, what time will it show at **8:00 PM** on the same day?

19. A clock gains **8 minutes in every 2 hours**. It is set correctly at **10:00 AM**. What will it read at **10:00 PM**?

20. A clock loses **6 minutes every 3 hours**. If it shows **3:00 PM** correctly, what will it show after **18 hours**?

21. A wall clock strikes once at **1 o'clock**, twice at **2 o'clock**, ..., and twelve times at **12 o'clock**. How many times will it strike in one complete day?

22. A clock takes **24 seconds** to strike **7 times**. How much time will it take to strike **11 times**?

23. A clock takes **18 seconds** to strike **10 times**. How long will it take to strike **16 times**?

24. A clock is placed on a table such that at **12:00 noon** the minute hand points towards the **North**. In which direction will the **hour hand** point at **3:00 PM**?

25. A clock is placed on a table such that at **6:00 PM** the hour hand points towards the **East**. In which direction will the **minute hand** point at **9:00 PM**?

Calendars:

1. What day of the week was **15th August 1947**?
2. What day of the week was **26th January 1950**?
3. If today is **Wednesday**, what day of the week will it be **95 days** from today?
4. If today is **Friday**, what day of the week was it **84 days ago**?
5. If **1st January 2026** is a Thursday, what day of the week will **31st December 2026** be?
6. A leap year has 366 days. What is the number of **odd days** in a leap year?
7. How many **odd days** are there in **400 consecutive years**?
8. Which of the following years is **not** a leap year?
    - 1900
    - 2000
    - 2024
    - 2400
9. If **8th December 2007** was a Saturday, what day of the week was **8th December 2006**?
10. The calendar of the year **2025** will be repeated next in which year?

11. The calendar of the leap year **2024** will be repeated next in which year?

12. If **15th March 2023** was a Wednesday, what day of the week was **15th March 2022**?

13. If **10th October 2018** was a Wednesday, what day of the week was **10th October 2025**?

14. If today is **Monday**, what day of the week will it be after $\mathbf{10^2 + 5^2}$ **days**?

15. The first day of a year is **Tuesday**. What day of the week will be the **last day** of the same (non-leap) year?

16. Which of the following years has the **same calendar** as **2019**?

- 2025
- 2030
- 2031
- 2032

17. How many leap years are there between **1901 and 2000** (both inclusive)?

18. If **Republic Day (26 January)** falls on a Sunday in a particular year, on which day will **Independence Day (15 August)** fall in the same year?

19. A century year is a leap year only if it is divisible by which of the following?

- 4
- 100
- 200
- 400

20. If **31st December 2027** is a Friday, what day of the week will **1st January 2028** be?

Syllogisms:

1.

Statements:

- All Roses are Flowers.
- All Flowers are Plants.

Conclusions:
I. All Roses are Plants.
II. All Plants are Roses.

2.

Statements:

- Some Engineers are Managers.
- All Managers are Leaders.

Conclusions:
I. Some Engineers are Leaders.
II. All Engineers are Leaders.

3.

Statements:

- No Doctor is a Lawyer.
- All Lawyers are Graduates.

Conclusions:
I. No Doctor is a Graduate.
II. Some Graduates are Lawyers.

4.

Statements:

- All Books are Papers.
- Some Papers are Files.

Conclusions:
I. Some Files are Books.
II. All Books are Files.

5.

Statements:

- Some Cars are Electric.
- All Electric vehicles are Silent.

Conclusions:
I. Some Cars are Silent.
II. All Cars are Silent.

6.

Statements:

- No Bird is a Mammal.
- Some Mammals are Pets.

Conclusions:
I. Some Pets are Mammals.
II. Some Birds are Pets.

7.

Statements:

- All Apples are Fruits.
- No Fruit is a Vegetable.

Conclusions:
I. No Apple is a Vegetable.
II. Some Vegetables are Apples.

8.

Statements:

- Some Pens are Markers.
- Some Markers are Highlighters.

Conclusions:
I. Some Pens are Highlighters.
II. Some Highlighters are Markers.

9.

Statements:

- Only Teachers are Professors.
- All Professors are Educated.

Conclusions:
I. All Professors are Teachers.
II. All Teachers are Educated.

10.

Statements:

- All Mobile Phones are Electronic Devices.
- Some Electronic Devices are Laptops.

Conclusions:
I. Some Laptops are Mobile Phones.
II. All Mobile Phones are Laptops.

---

11. ***(Possibility Based)***

Statements:

- All Engineers are Employees.
- Some Employees are Managers.

Conclusions:
I. Some Engineers being Managers is a possibility.
II. All Managers being Engineers is a possibility.

---

12. ***(Possibility Based)***

Statements:

- No Cats are Dogs.
- Some Dogs are Pets.

Conclusions:
I. Some Cats being Pets is a possibility.
II. Some Pets are Dogs.

13.

Statements:

- Some Students are Athletes.
- All Athletes are Fit.
- No Fit person is Lazy.

Conclusions:
I. Some Students are not Lazy.
II. No Athlete is Lazy.

14.

Statements:

- All Squares are Rectangles.
- All Rectangles are Parallelograms.
- Some Parallelograms are Rhombuses.

Conclusions:
I. All Squares are Parallelograms.
II. Some Rhombuses are Rectangles.

15.

Statements:

- Some Computers are Laptops.
- All Laptops are Portable.
- No Portable device is Heavy.

Conclusions:
I. Some Computers are not Heavy.
II. No Laptop is Heavy.
III. Some Heavy objects are Computers.

Missing Number Series

1. Find the missing number:
   3, 10, 21, 36, ?, 78, 105
2. Find the missing number:
   8, 17, 36, 75, ?, 307
3. Find the missing number:
   5, 9, 17, 33, ?, 129
4. Find the missing number:
   2, 6, 18, 54, ?, 486
5. Find the missing number:
   13, 18, 28, 43, ?, 88
6. Find the missing number:
   1, 4, 10, 22, ?, 94

Find the Next Number

7. Find the next number:
   6, 11, 21, 41, 81, ?
8. Find the next number:
   2, 5, 11, 23, 47, ?
9. Find the next number:
   4, 9, 19, 39, 79, ?

10. Find the next number:
    1, 2, 6, 24, 120, ?

11. Find the next number:
    10, 18, 34, 66, 130, ?

12. Find the next number:
    7, 14, 12, 24, 22, 44, ?

13. Find the next number:
    3, 7, 15, 31, 63, ?

Find the Wrong Number

14. Find the wrong number in the series:
    2, 6, 12, 20, 32, 46, 56

15. Find the wrong number:
    1, 8, 27, 64, 124, 216, 343

16. Find the wrong number:
    5, 11, 23, 47, 95, 191, 383

17. Find the wrong number:
    4, 9, 16, 24, 36, 49, 64

18. Find the wrong number:
    3, 6, 12, 24, 49, 96, 192

Alternating Series

19. Find the missing number:
    2, 5, 6, 11, 18, 17, 54, ?

20. Find the next number:
    100, 95, 90, 82, 74, 61, ?

21. Find the next number:
    9, 27, 30, 90, 93, 279, ?

Mixed Pattern Series

22. Find the next number:
    1, 3, 7, 15, 31, 63, ?

23. Find the next number:
    2, 3, 7, 25, 121, ?

24. Find the next term:
    A, C, F, J, O, ?

25. Find the missing term:
    Z, W, S, N, H, ?, ?

1. Linear Arrangement

Eight friends **A, B, C, D, E, F, G, and H** are sitting in a row facing North.

- A sits third to the left of D.
- B is an immediate neighbour of D.
- C sits at one of the ends.
- E sits second to the right of B.
- F sits immediately to the left of C.
- G is not adjacent to A.
- H sits between G and A.

Who sits exactly in the middle of the row?

---

2. Circular Arrangement (Facing Centre)

Seven friends **P, Q, R, S, T, U, and V** are sitting around a circular table facing the centre.

- P is second to the left of Q.
- R sits opposite U.
- S is an immediate neighbour of P and T.
- V sits second to the right of R.

Who sits immediately to the left of U?

---

3. Linear Arrangement

Six students **K, L, M, N, O, and P** are sitting in a row facing South.

- M sits at one of the ends.
- O is second to the left of P.
- L is between K and N.
- N is an immediate neighbour of O.

Who sits at the other end?

---

4. Square Arrangement

Four persons **A, B, C, and D** are sitting at the four corners of a square facing the centre.

- A sits opposite C.
- B is to the immediate left of A.

Who is to the immediate right of C?

---

5. Circular Arrangement (Facing Outside)

Eight people **A, B, C, D, E, F, G, and H** are sitting around a circular table facing outward.

- A sits second to the right of B.
- C is opposite F.
- G is between B and H.
- D is an immediate neighbour of F.
- E is second to the left of D.

Who sits immediately to the left of A?

---

6. Parallel Rows

Eight people are sitting in two parallel rows with four people in each row.

- Row 1 (facing South): A, B, C, D
- Row 2 (facing North): P, Q, R, S

Additional information:

- A sits opposite Q.
- B is to the immediate left of A.
- C sits opposite S.
- P is at one of the ends.

Who sits opposite B?

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7. Rectangular Table

Eight friends are sitting around a rectangular table. Four sit at the corners facing the centre, while four sit at the middle of each side facing outward.

- A sits at a corner.
- B sits opposite A.
- C sits between A and D.
- E sits immediately to the right of B.
- F sits opposite C.
- G is adjacent to E.
- H sits between G and A.

Who sits opposite H?

---

8. Linear Arrangement

Nine persons **A, B, C, D, E, F, G, H, and I** are sitting in a row facing North.

- Only three people sit between A and D.
- B sits immediately to the right of D.
- E sits second to the left of A.
- G is at one of the ends.
- H sits between I and F.
- C is not adjacent to D.

Who sits third from the right end?

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9. Circular Arrangement with Mixed Facing

Eight friends sit around a circular table.

- A, C, E, and G face the centre.
- B, D, F, and H face outward.
- A sits opposite E.
- B sits second to the left of A.
- C sits immediately to the right of D.
- F sits opposite G.
- H is an immediate neighbour of B.

Who sits immediately to the left of G?

---

10. Puzzle-Based Seating Arrangement

Ten friends **A, B, C, D, E, F, G, H, I, and J** are sitting in a straight line facing North.

- A sits fourth to the left of J.
- Exactly two persons sit between B and A.
- C is an immediate neighbour of J.
- D sits third to the right of B.
- E sits at one of the ends.
- F sits immediately to the left of G.
- H is not adjacent to F.
- I sits between H and D.

Who sits fifth from the left end?